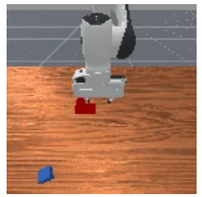
4D MAE / B5

PullCube configuration

(a) Policy architecture



Current RGB

ResNet18
RGB stem + resampler

16 tokens

Robot state

State MLP
+ resampler

16 tokens

Point history C

Tokenizer Tθ
+ point resampler

16 tokens

Learned queries
64 action + 16 future

HPT
128 × 256
16 blocks
8 heads

Concat. + position

64 action

Noisy actions + τ

Action flow head
6 blocks · width 128

Action chunk
32 × 8
Execute 8; replan

16 future-context tokens (K, V)

Point query Q
Current XYZ

Tθ
Point token

q

Trajectory decoder
q attends to K, V

Noisy tracks + flags

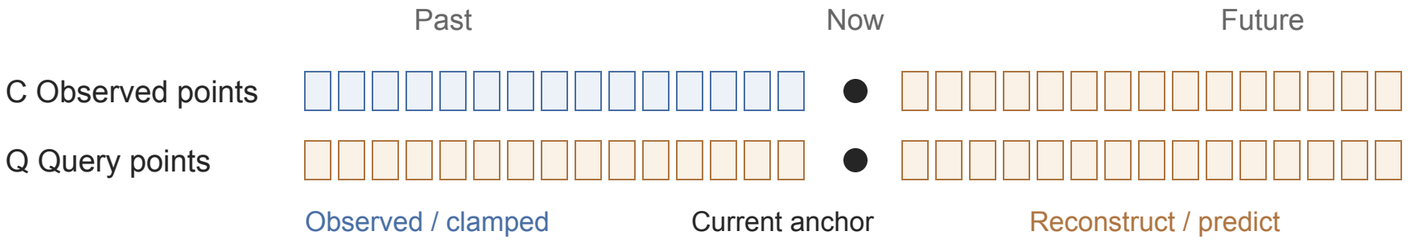
Flow time τ

Q completion
50 flow steps
15 past + 15 future

Tθ: shared weights

Control uses the action branch. Trajectory completion supplies auxiliary training supervision.

(b) Trajectory supervision



$$L = L_{action} + (L_{C,future} + L_{Q,past} + L_{Q,future}) / 3$$

Flow-matching losses on valid targets.

C: same decoder; observed past stays fixed.

64 C points · 64 Q points · 16 history frames · privileged simulator history in PullCube

All trajectory sketches are schematic.

Inside HPT

B5 / PullCube · 128 tokens × 256 dimensions

Heterogeneous Pretrained Transformer: modality-specific stems → shared transformer → task readouts

Observation-dependent features

RGB stem
16 × 256

State stem
16 × 256

History tokenizer + resampler
16 × 256 · B5 extension

Learned input tokens · EgoWAM

64 action tokens
Learned parameter vectors

16 future tokens
Learned parameter vectors

Concat.
+ position

Shared transformer trunk

16 blocks · width 256 · 8 attention heads

LayerNorm → self-attention
Residual addition

LayerNorm → MLP
Residual addition

All token groups attend to each other.

One forward pass per observation.

No noisy trajectories or flow time τ here.

Read out the corresponding positions

64 action-context tokens
Condition the action flow head

16 future-context tokens
Condition the trajectory decoder

Both outputs depend on this observation.

Token counts are latent capacity;
they do not count predicted time steps.

Two separate inputs to the trajectory decoder

HPT context is fixed during 50 generation steps

16 future-context tokens
From HPT

Condition via cross-attention

Trajectory decoder
Predict a velocity field

Update
the trajectory

Flow time τ

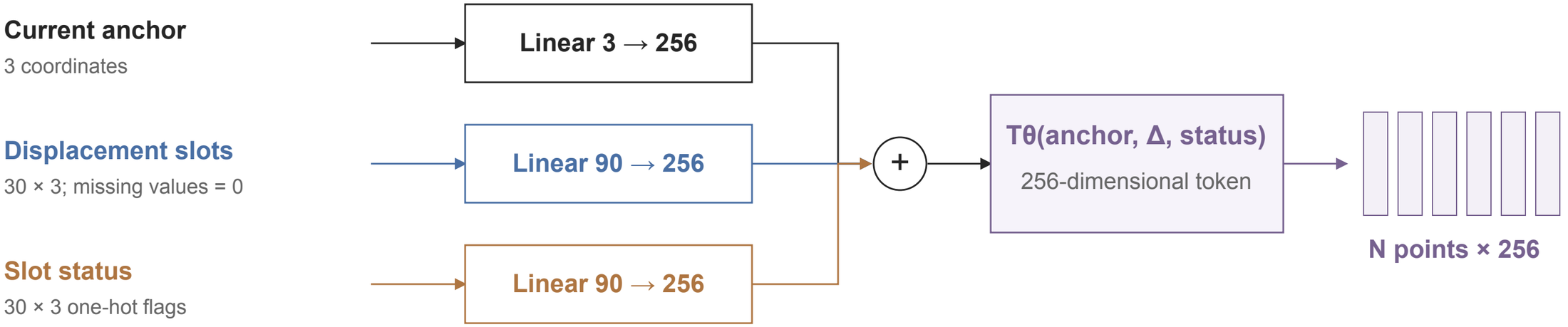
Noisy displacements + anchors
+ observed / noisy slot flags

Tokenizer Tθ
One token per point

The 16 HPT context tokens bypass Tθ. The decoder combines context with the evolving trajectory.

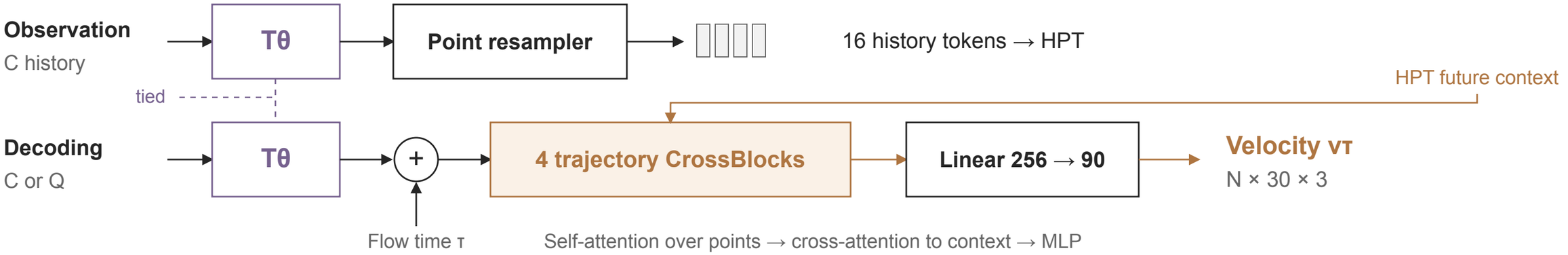
B5: shared trajectory tokenizer

(a) One trajectory → one point token



Status: missing / observed / noisy. All 30 time slots are packed into each point token.

(b) The same Tθ in both paths



Sampling: 50 Euler steps. Observed C past is clamped at every step.

τ is generative time; the 30 output slots are physical time. Add the current anchor to recover absolute XYZ.